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SECOND SEMESTER M.Tech EXAMINATION, JUNE 2024

INTRODUCTION TO FINITE ELEMENT METHOD

Time: 3 Hours **Maximum Marks: 100**

Instructions: **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 20 marks.

ii. Missing data may be suitably assumed.

1.	**a.**	**i.** How do you characterize the stress state of structures which are loaded in their plane. Write the equilibrium, strain displacement and constitutive equation for such a state of stress. Give Examples **ii.** Explain the salient features of Axisymmetric modelling of structures with concrete examples	CO1	L1	**6+4=10**
	b.	**i.** State the principal of minimum potential energy and principle of minimum complimentary energy principle. Give their mathematical statements **ii.** State what conditions are derivable from the minimization of the corresponding functional from the principal of minimum potential energy and principle of minimum complimentary energy.	CO1	L1	**5+5=10**
2.	**a.**	Explain the four primary quantities required to be assessed using Finite element Analysis? Discuss why are these primary quantities important in design	CO2	L2	**10**
	b.	Explain, the help of FEM in the Product Design. Explain the advantages of FEM over other numerical method	CO2	L2	**10**
3.	**a.**	Explain any three weighted residual methods for developing approximate solutions. Discuss the advantages and disadvantages of each method over each other	CO2	L2	**10**
	b.	A Functional which is dependent on several dependent variables $q_1, q_2, ..q_n$ and independent variable 't' $$S=\int_{x_1}^{x_2} f[q_1,q_2,q_3,...\dot{q}_1,\dot{q}_2....\dot{q}_n,t]$$	CO2	L2	**10**

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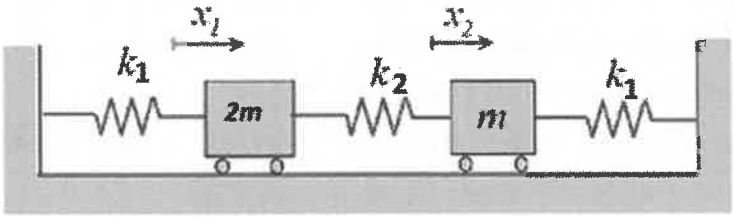
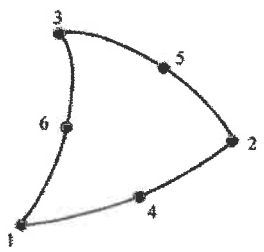
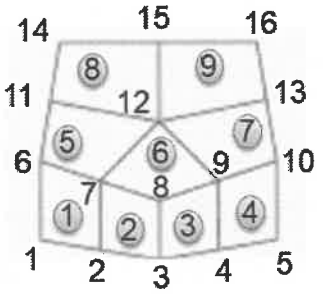
		The Euler Lagrange equations is given by $\left(\frac{\partial F}{\partial q_i} - \frac{d}{dt}\frac{\partial F}{\partial \dot{q}_i}\right) = 0 \quad i = 1,2,\ldots,n$ Using this establish the governing equations of the following system, obtain the governing equation.  Hint (Use Principle of Minimum Potential Energy)			
		OR			
4.	**a.**	**i.** Write the difference between principle of virtual displacements and principle of virtual forces. **ii.** State the Castigliano first and second theorems. **iii.** Explain why Finite elements are developed using principle of virtual displacements and not principle of virtual forces	CO3	L1	**5+2+3= 10**
	b.	**i.** Obtain the weak form of the following Governing differential Equations. $AE\frac{d^2u}{dx^2} + q_0 = 0$ **ii.** Identify the primary variable and the secondary variable from the weak statement. **iii.** Identify all possible conditions of the problem and classify them as Kinematic or Natural Boundary condition.	CO3	L2	**6+2+2= 10**
5.	**a.**	**i.** Express the need of Numerical Integration in FE? **ii.** Explain the two methods of Numerical Integration used in FE. **iii.** Explain the differences between these two methods and discuss which method is more popular? Give reasons for your answer.	CO4	L2	**3+2+5= 10**
	b.	**i.** Obtain the solution by Numerically Integrate $I_\xi = \int_{-1}^{1}(3\xi^2 - 2\xi + 10)d\xi$ using Newton cotes Quadrature using 1-, 2- and 3-Point rule.	CO4	L4	**5+2+3= 10**

		ii. Evaluate the exact Integration and compare them with Numerical integration **iii.** Compare Results and draw your conclusions. The following table can be used for Newton Cotes Quadrature **Weighting coefficients for the Newton–Cotes formula**			

r	W_1	W_2	W_3	W_4	W_5	W_6
1	1					
2	$\frac{1}{2}$	$\frac{1}{2}$				
3	$\frac{1}{6}$	$\frac{4}{6}$	$\frac{1}{6}$			
4	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$		
5	$\frac{7}{90}$	$\frac{32}{90}$	$\frac{12}{90}$	$\frac{32}{90}$	$\frac{7}{90}$	
6	$\frac{19}{288}$	$\frac{75}{288}$	$\frac{50}{288}$	$\frac{50}{288}$	$\frac{75}{288}$	$\frac{19}{288}$
7	$\frac{41}{840}$	$\frac{216}{840}$	$\frac{27}{840}$	$\frac{272}{840}$	$\frac{27}{840}$	$\frac{216}{840}$

		OR			
6.	**a.**	**i.** Distinguish between Natural Coordinates and global coordinate system. **ii.** Explain how do you define Natural coordinate system in 1D and 2D coordinate system. **iii.** What is the utility of natural coordinates in finite element analysis? Explain **iv.** What are the limitations of using Natural coordinates	CO4	L2	**4+2+2+2=10**
	b.	**i.** Define Area Coordinates for a Traingle.? **ii.** Explain why only two of the Area Coordinates are only independent. **iii.** Derive the interpolation functions for a six noded triangular element shown here using independent Area coordinates L_1, L_2  	CO4	L3	**2+1+7=10**
7.	**a.**	**i.** What is meant by Convergence. Differentiate between h-convergence and p-convergence. **ii.** Discuss the meaning of Completeness	CO5	L4	**4+3+3=10**

		Requirement in Finite element analysis **iii.** Explain what do we mean by geometric invariance. Wh is it important			
	b.	**i.** For the Mesh shown in the Figure, define the Connectivity Matrix. **ii.** Why is Jacobian check performed in Finite element Analysis. **iii.** Explain why Non Convex elements are not admissible in Finite element Analysis. **iv.** What is the ideal interior angle needed to be maintained in an element while meshing.  Derive Jacobian matrix for Rectangular element with the (x,y) co-ordinates of the nodes are (1.5,2) , (7,3.5) (10,8.5) and (4,7) at nodes I,j,k,l	CO5	L1	**5+2+2+1=10**
		OR			
8.	**a.**	**i.** Write the difference between C0 Continuity and C1 Continuity. **ii.** Give Examples of C^0 and C^1 Continuity Problems **iii.** How do we distinguish if a FE formulation requires C^0 or C^1 Continuity. **iv.** How do we choose Interpolation Functions for C^0 and C^1 continuity Problems and establish the Interpolating polynomials for a typical 1D C1 Continuous Problem.	CO6	L2	**2+2+2+4=10**
	b.	Given a Global Stiffness Matrix of Size 8 in which the known displacements are $\delta_2, \delta_5, and\ \delta_7$ write the detailed procedure for the imposition of Boundary conditions, Determination of Secondary Variables and subsequent Solution	CO6	L2	**10**

$$[K] = \begin{bmatrix} K_{11} & K_{12} & K_{13} & K_{14} & K_{15} & K_{16} & K_{17} & K_{18} \\ K_{12} & K_{22} & K_{23} & K_{24} & K_{25} & K_{26} & K_{27} & K_{28} \\ K_{13} & K_{23} & K_{33} & K_{34} & K_{35} & K_{36} & K_{37} & K_{38} \\ K_{14} & K_{24} & K_{34} & K_{44} & K_{45} & K_{46} & K_{47} & K_{48} \\ K_{15} & K_{25} & K_{35} & K_{45} & K_{55} & K_{56} & K_{57} & K_{58} \\ K_{16} & K_{26} & K_{36} & K_{46} & K_{56} & K_{66} & K_{67} & K_{68} \\ K_{17} & K_{27} & K_{37} & K_{47} & K_{57} & K_{67} & K_{77} & K_{78} \\ K_{18} & K_{28} & K_{38} & K_{48} & K_{58} & K_{68} & K_{78} & K_{88} \end{bmatrix} \begin{Bmatrix} \delta_1 \\ \delta_2 \\ \delta_3 \\ \delta_4 \\ \delta_5 \\ \delta_6 \\ \delta_7 \\ \delta_8 \end{Bmatrix}$$

$$= \begin{Bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \\ f_5 \\ f_6 \\ f_7 \\ f_8 \end{Bmatrix} + \begin{Bmatrix} Q_1 \\ Q_2 \\ Q_3 \\ Q_4 \\ Q_5 \\ Q_6 \\ Q_7 \\ Q_8 \end{Bmatrix}$$

Where $\{f_i\}$ and $\{Q_i\}$ are the load vector due to external loads and secondary variable vector. Formulate your Answers in terms of Matrix Notation.